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Unit 3 : Session 10 : SLO 1 — Tutorial: Restoring Division

■ Problem 1: Restoring Division of 0111 (7) ÷ 0011 (3)

- Init: $A = 0000$, $Q = 0111$, $M = 0011$
- Step 1: shift (A,Q) → $A = 0000$, $Q = 1110$; $A = A - M = 0000 - 0011 = 1101 (< 0)$ → restore $A = 0000$, $Q_0 = 0$ → $A = 0000$, $Q = 1110$
- Step 2: shift → $A = 0001$, $Q = 1100$; $A = A - M = 0001 - 0011 = 1110 (< 0)$ → restore $A = 0001$, $Q_0 = 0$ → $A = 0001$, $Q = 1100$
- Step 3: shift → $A = 0011$, $Q = 1000$; $A = A - M = 0011 - 0011 = 0000 (\geq 0)$ → $Q_0 = 1$ → $A = 0000$, $Q = 1001$
- Step 4: shift → $A = 0001$, $Q = 0010$; $A = A - M = 0001 - 0011 = 1110 (< 0)$ → restore $A = 0001$, $Q_0 = 0$ → $A = 0001$, $Q = 0010$
- Final Quotient = $Q = 0010$ (binary) = 2 (decimal)
- Final Remainder = $A = 0001$ (binary) = 1 (decimal)
- Verification: $7 \div 3 = 2$ remainder 1 ✓